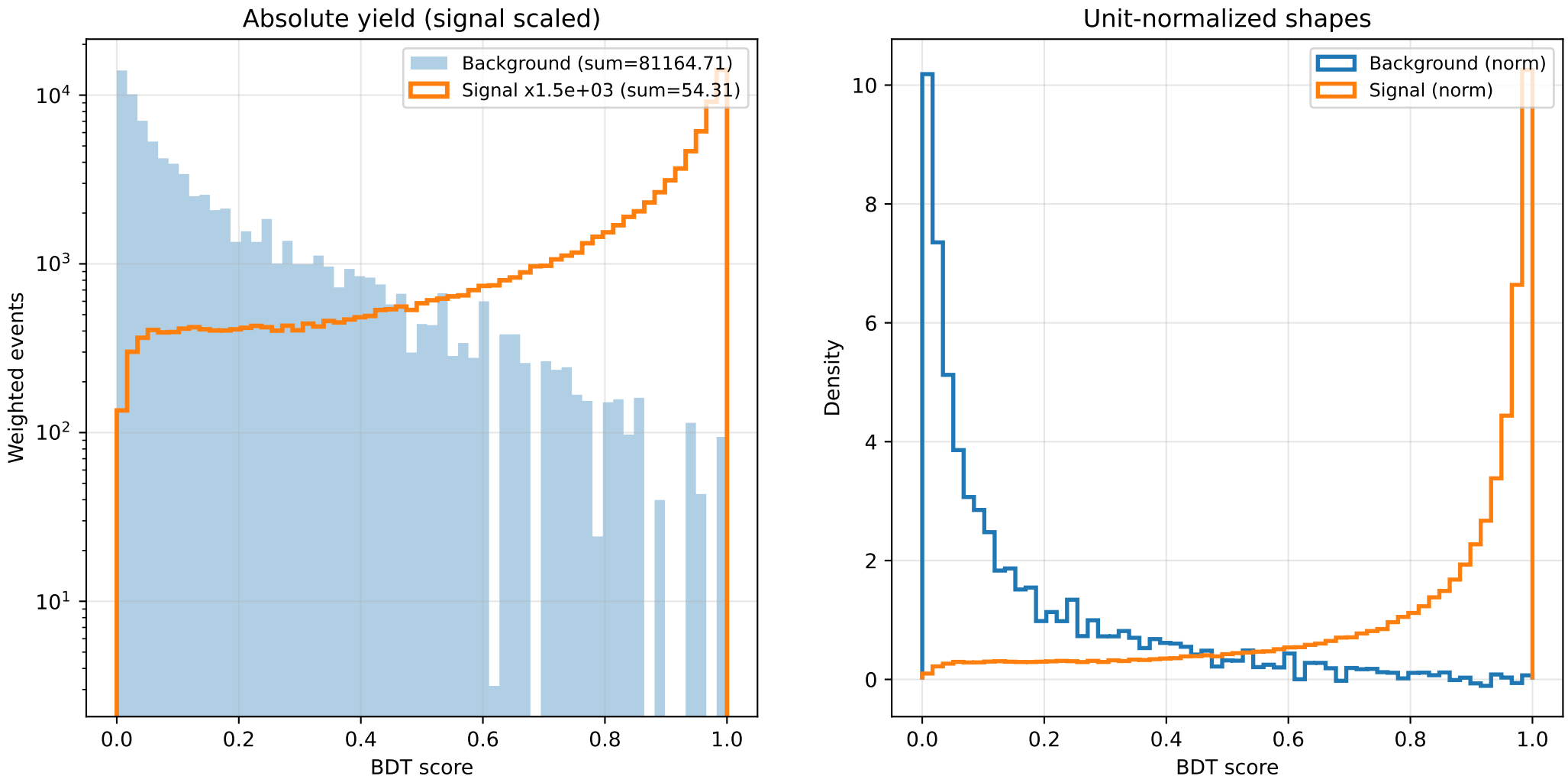
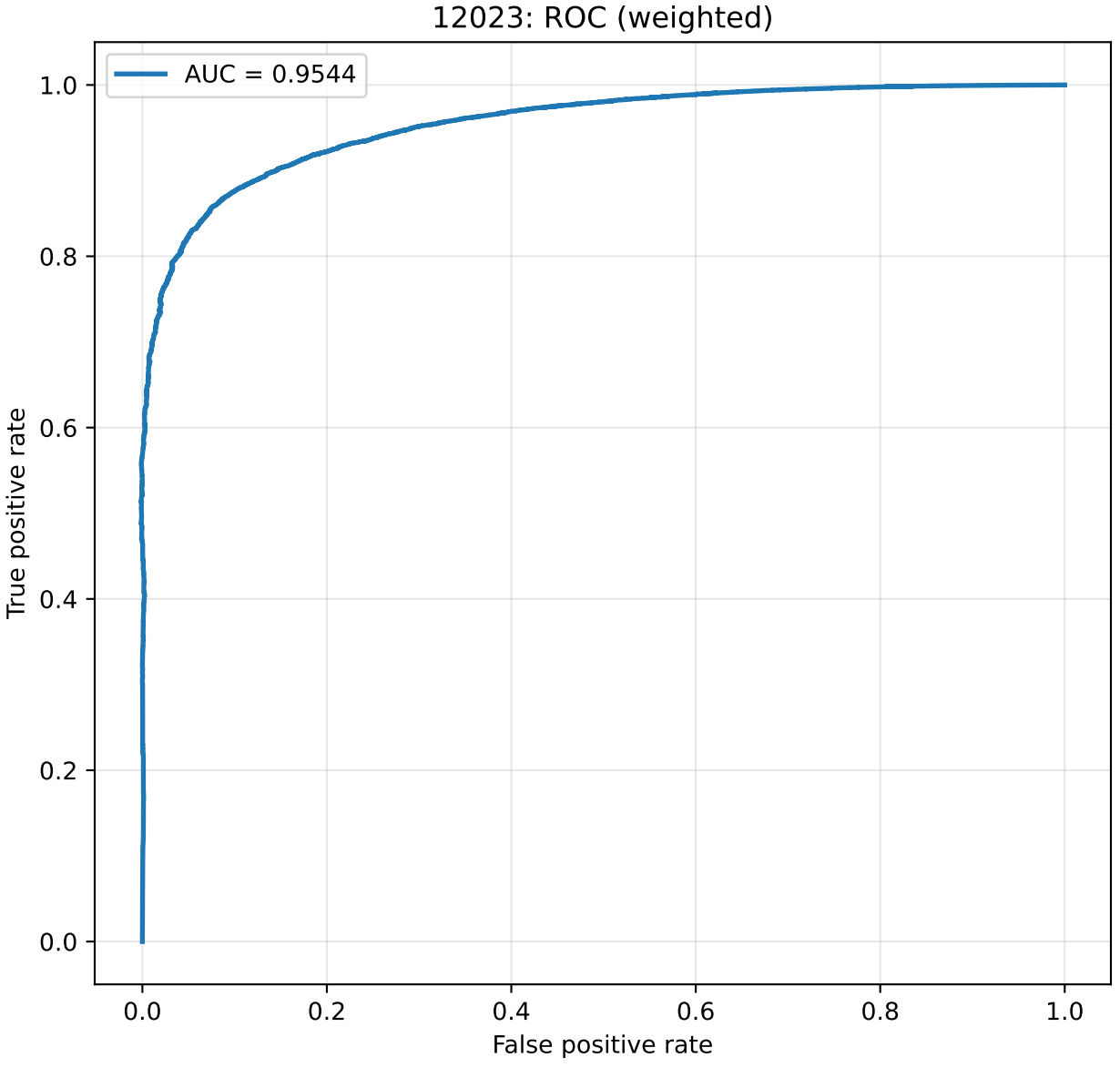
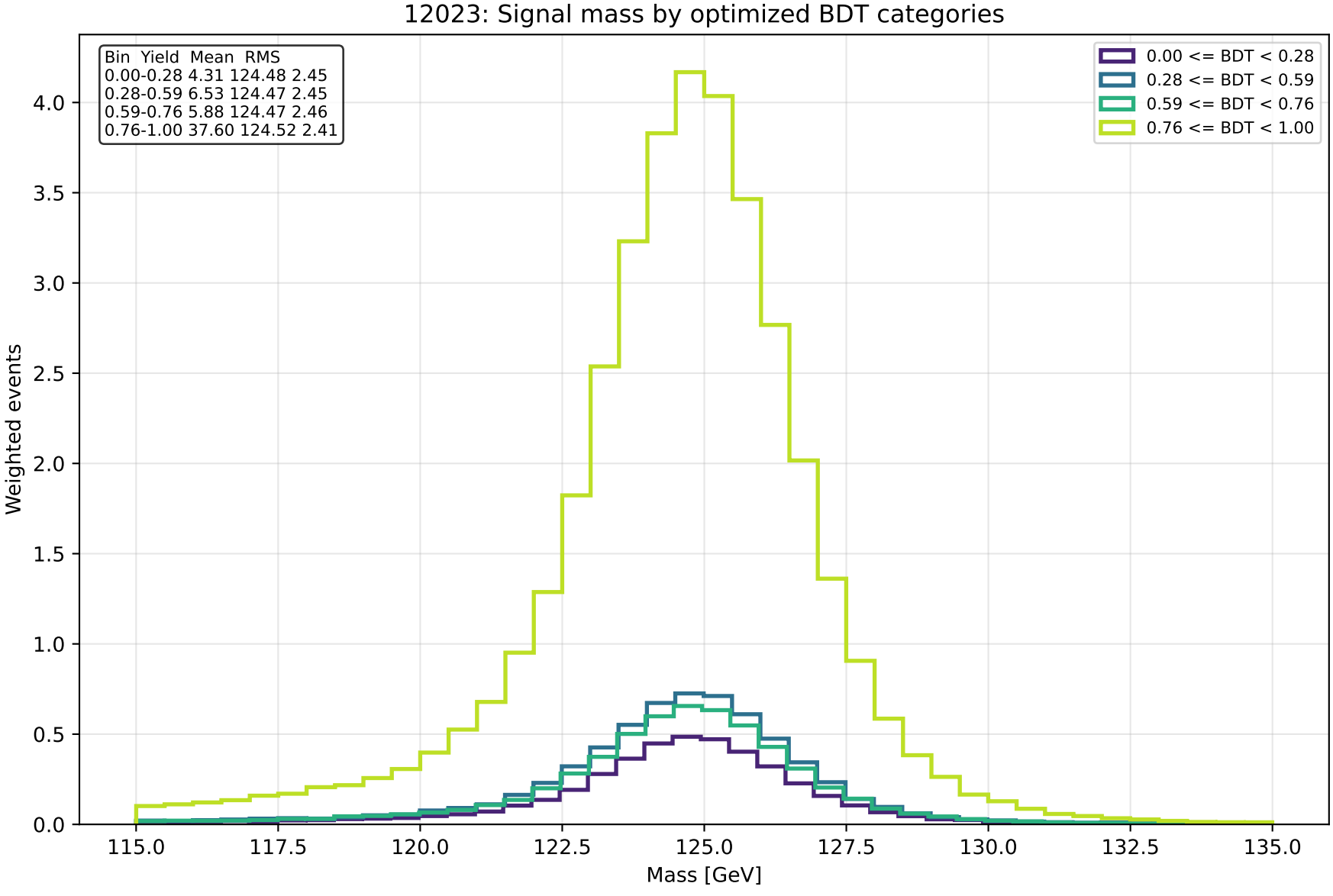
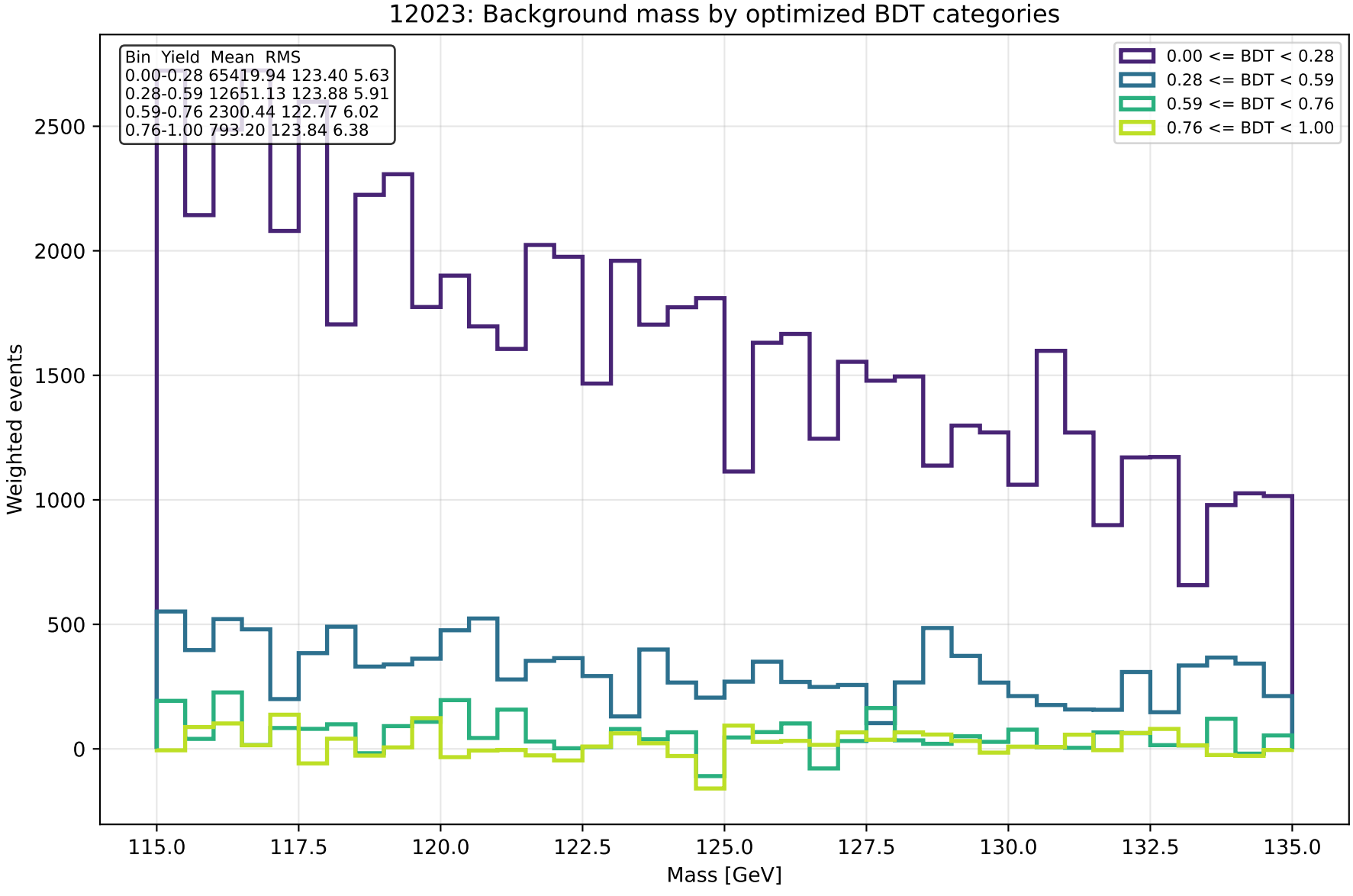

12023: BDT scores in 115-135 GeV
Absolute yield (signal scaled)
Background (sum=81164.71)
Signal x1.5e+03 (sum=54.31)
Weighted events
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
BDT score

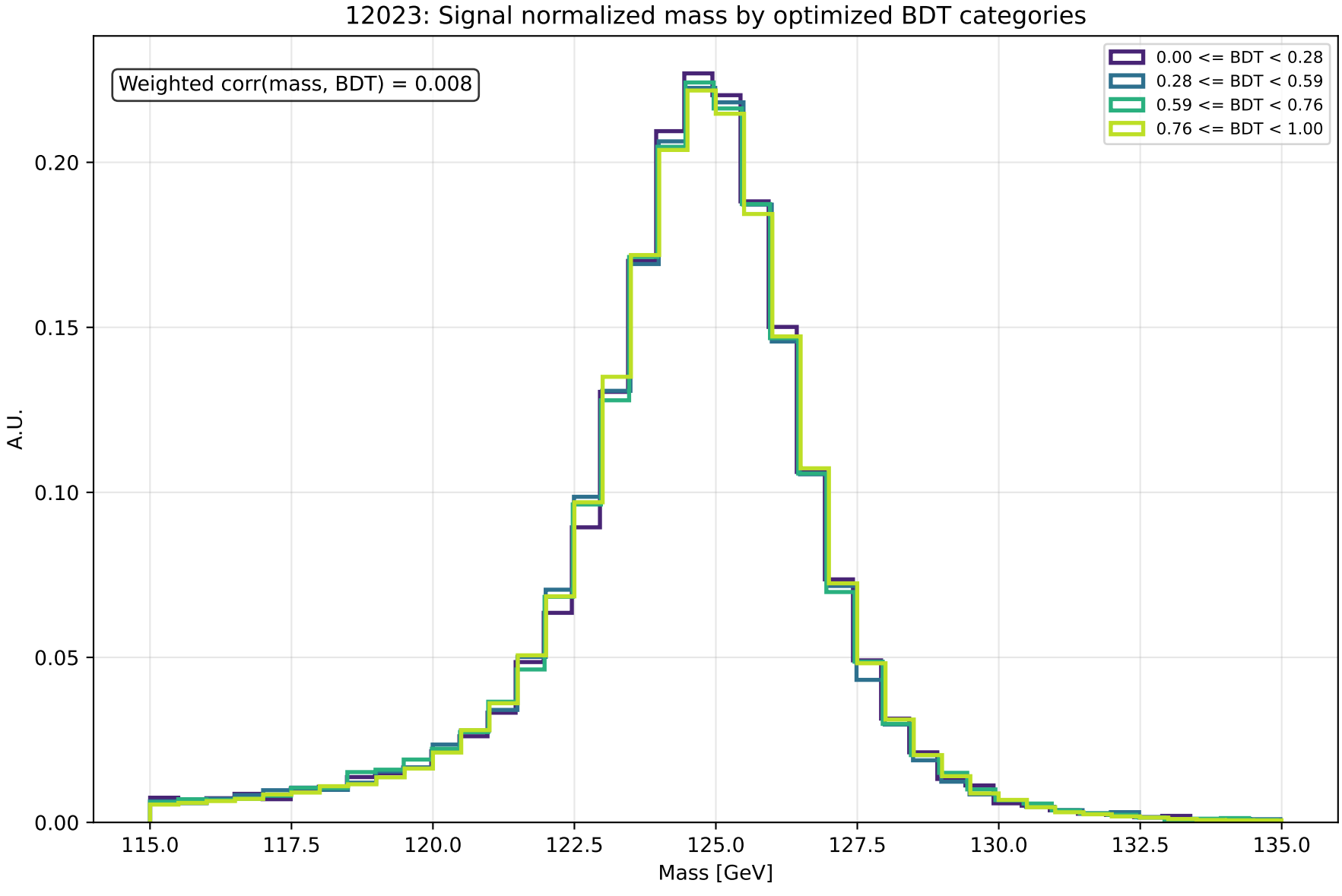

12023: ROC (weighted)
AUC = 0.9544
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

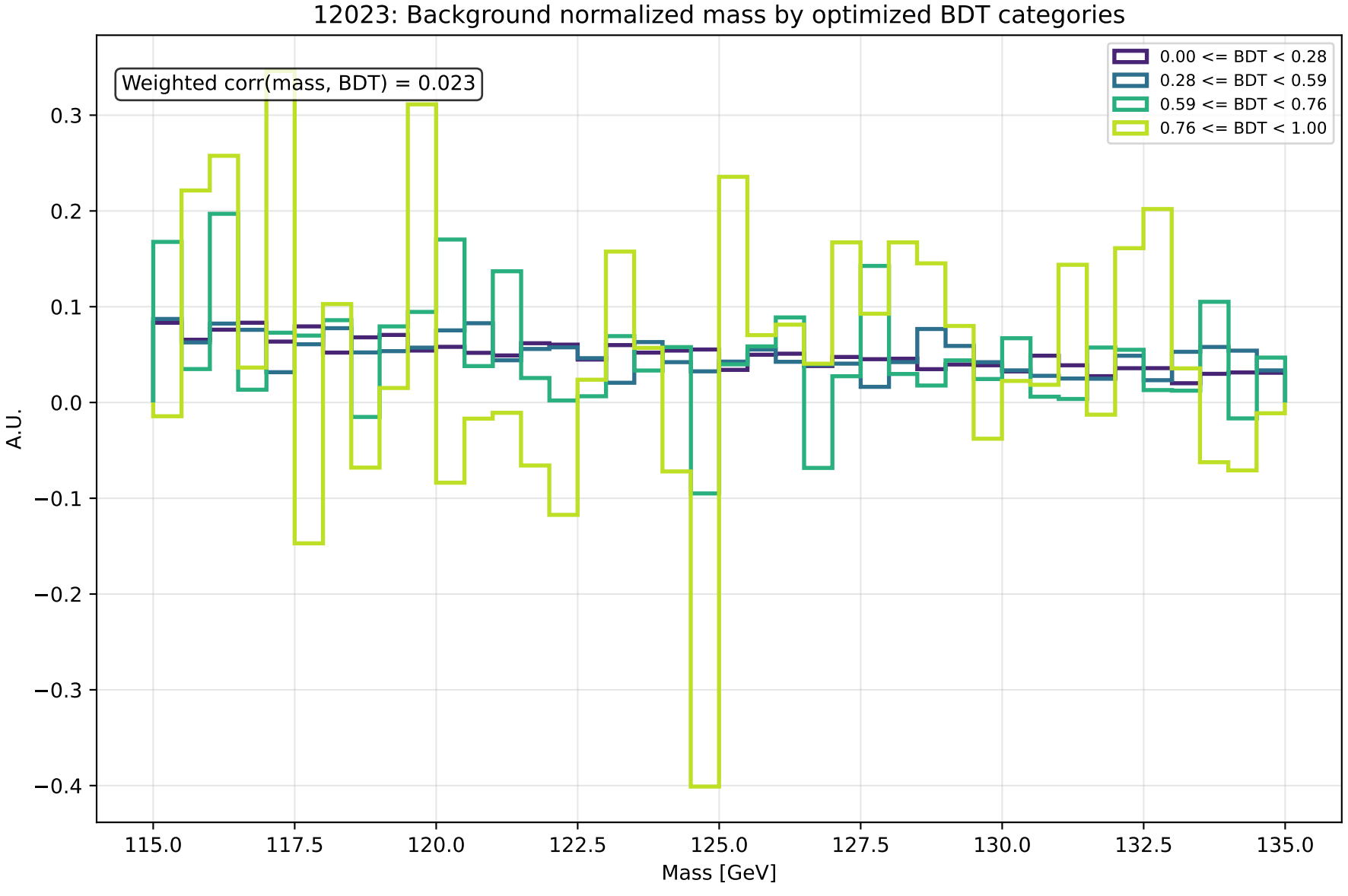
12023: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=4.051, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.283	4.308	65419.940	6.58e-05	0.017
2	0.283-0.594	6.529	12651.131	0.000516	0.058
3	0.594-0.756	5.876	2300.436	0.00255	0.123
4	0.756-1.000	37.600	793.205	0.0474	1.335


12023: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.28 4.31 124.48 2.45
0.28-0.59 6.53 124.47 2.45
0.59-0.76 5.88 124.47 2.46
0.76-1.00 37.60 124.52 2.41
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


12023: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.28 65419.94 123.40 5.63
0.28-0.59 12651.13 123.88 5.91
0.59-0.76 2300.44 122.77 6.02
0.76-1.00 793.20 123.84 6.38
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
2500
2000
1500
1000
500
0
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


12023: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.008
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

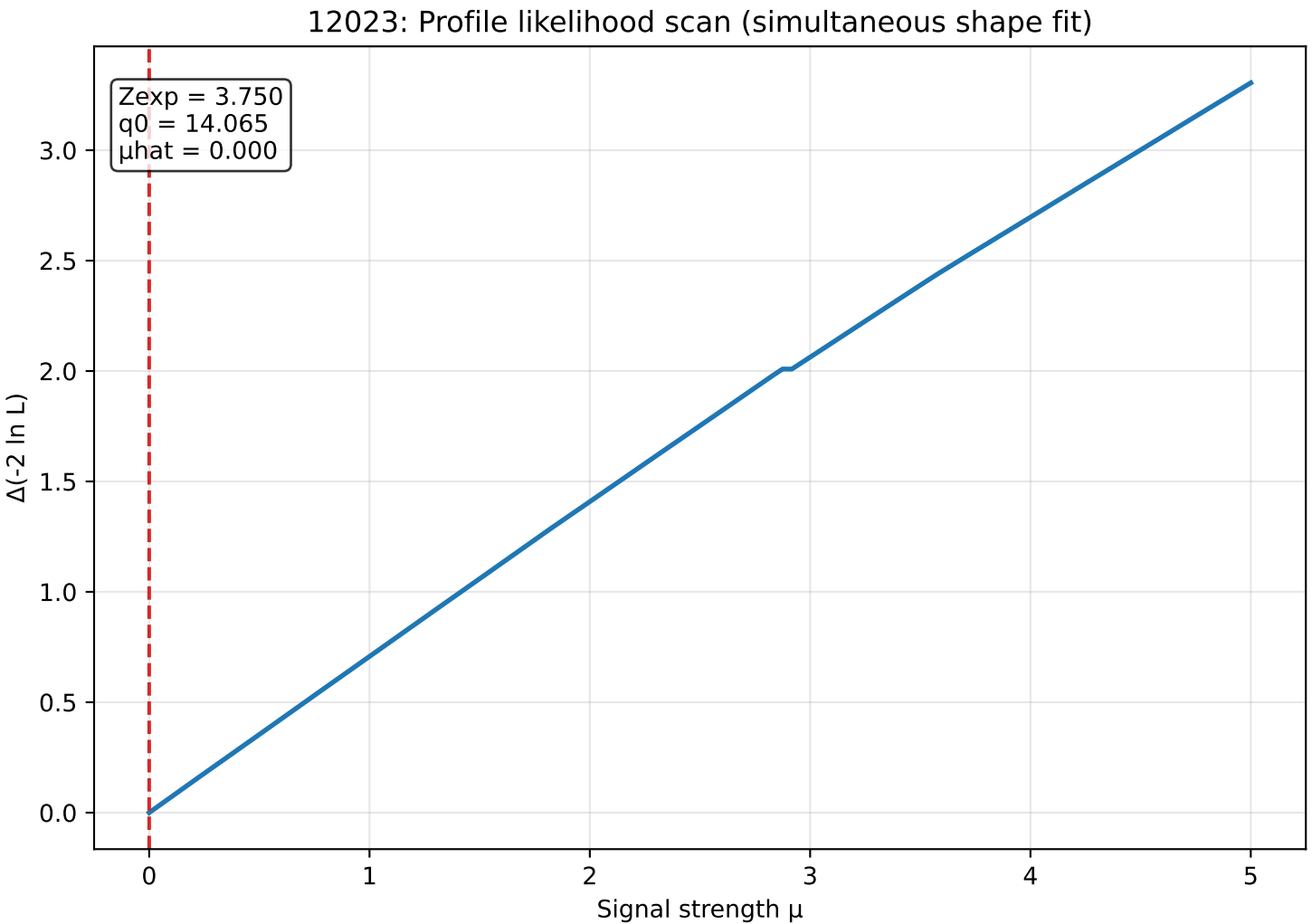

12023: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.023
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

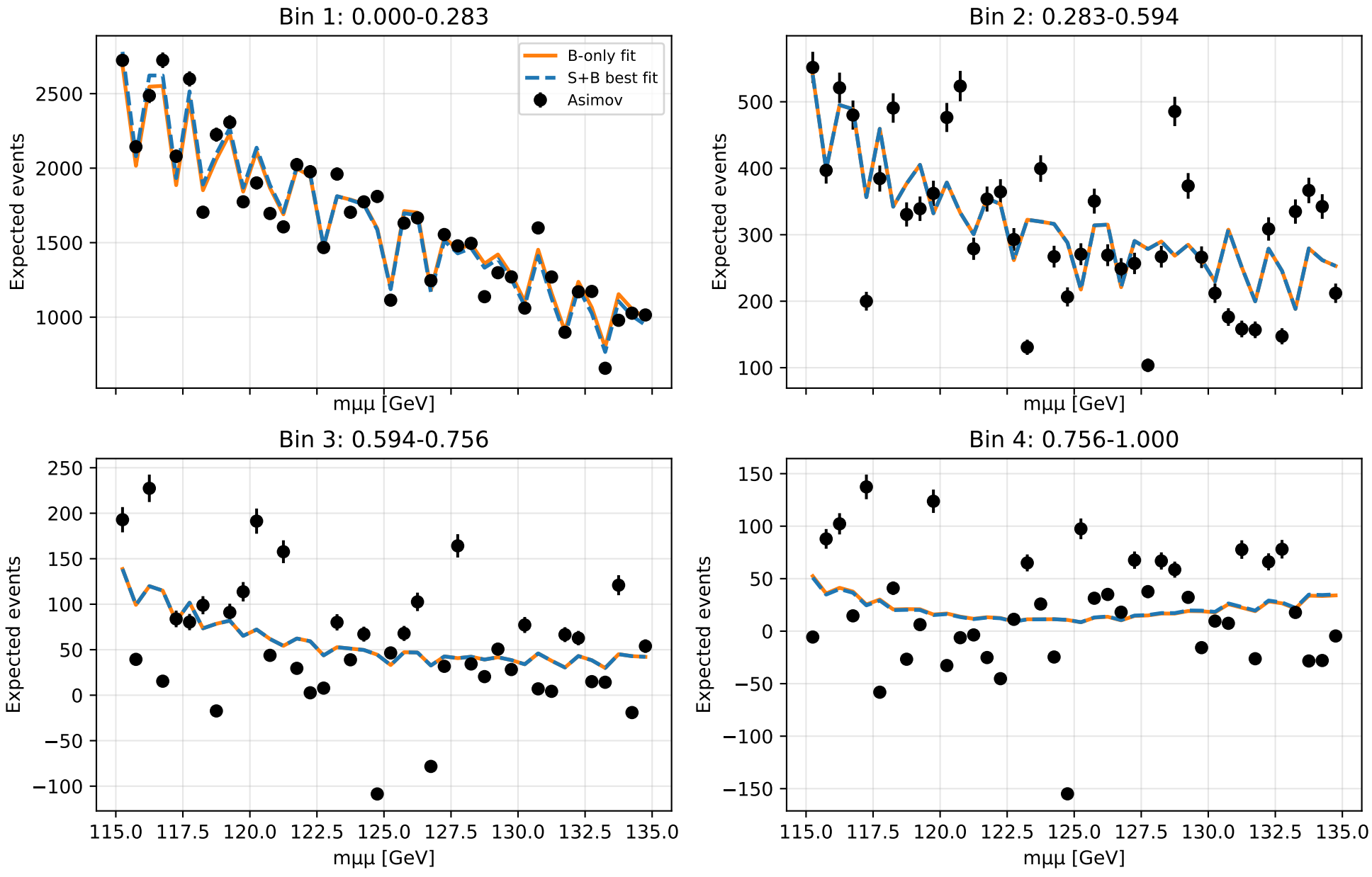
12023: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	3.7504
q0	14.0653
mu_hat	0.0000
N_BDT bins	4
N_mass bins	40
poly order	2
bkg norm unc	0.200
poly unc	0.300
core poly order	2
core from low/med bins	True
DY >=2j unc	0.050
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

12023: Shape-likelihood bin yields (Asimov)

Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.283	4.308	65419.940	65439.390	65439.390
2	0.283-0.594	6.529	12651.131	12654.711	12654.711
3	0.594-0.756	5.876	2300.436	2300.975	2300.975
4	0.756-1.000	37.600	793.205	832.605	832.605

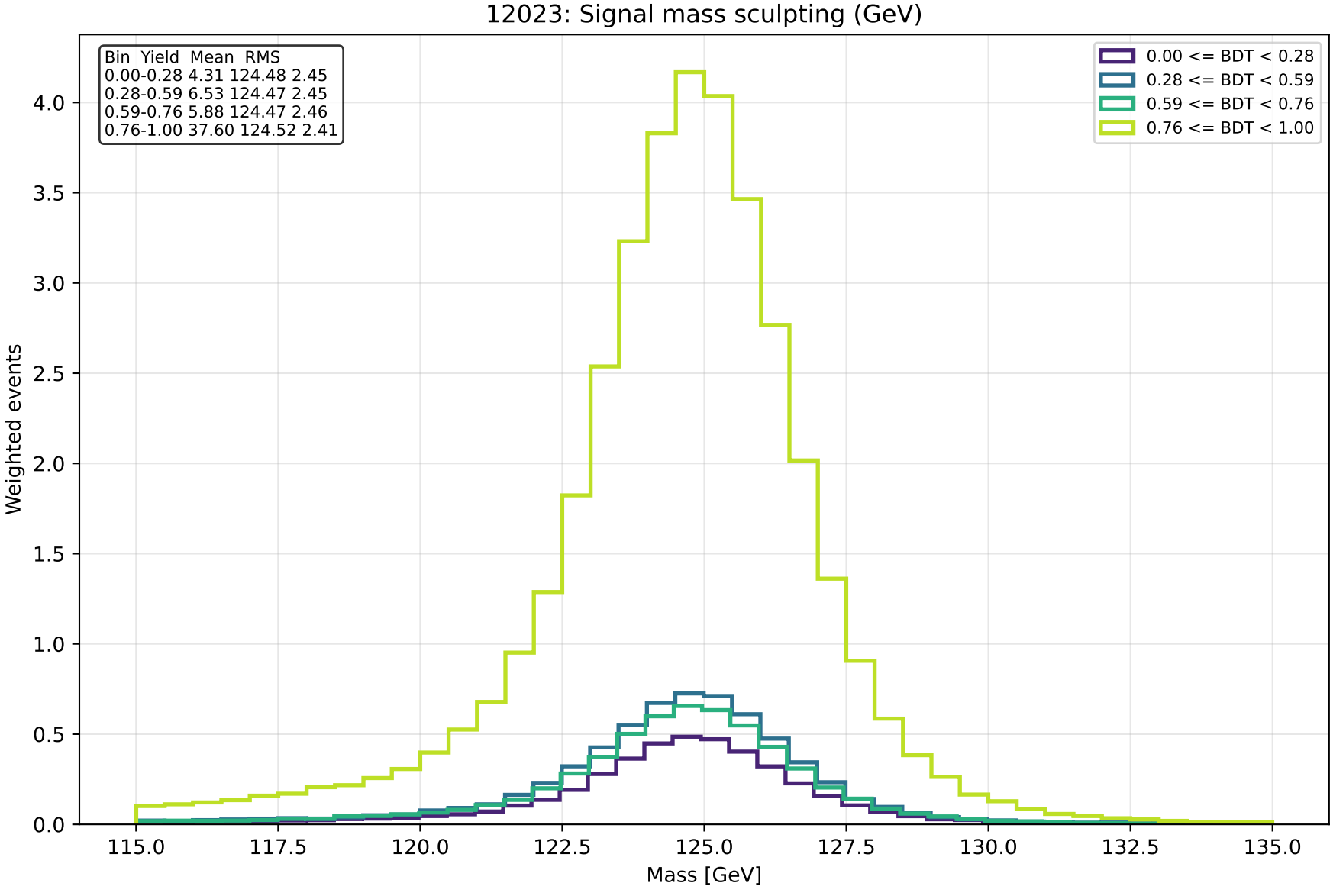

12023: Profile likelihood scan (simultaneous shape fit)
Zexp = 3.750
q0 = 14.065
μhat = 0.000
Δ(-2 ln L)
Signal strength μ
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
1
2
3
4
5

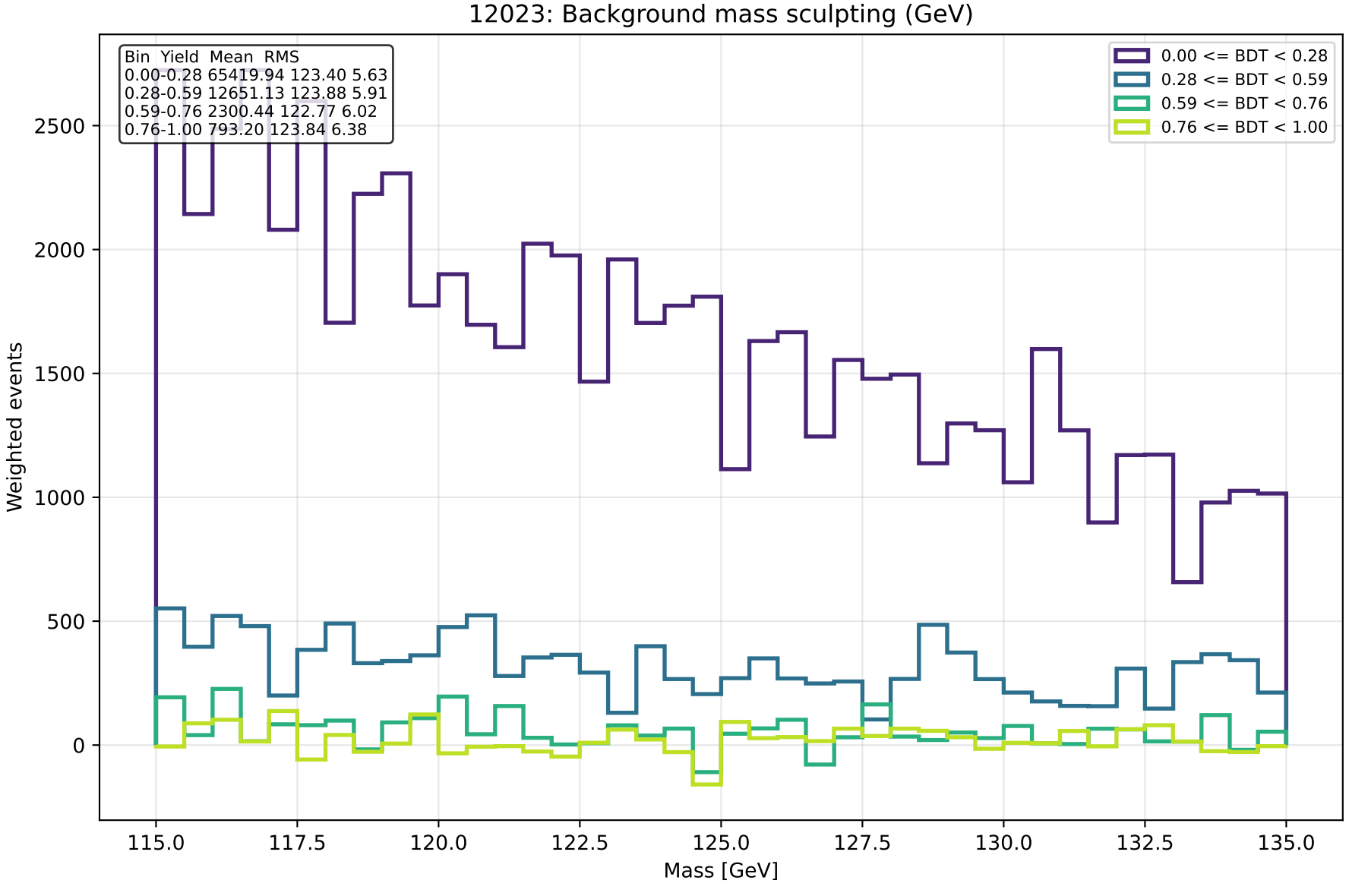

12023: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.283
Bin 2: 0.283-0.594
Bin 3: 0.594-0.756
Bin 4: 0.756-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

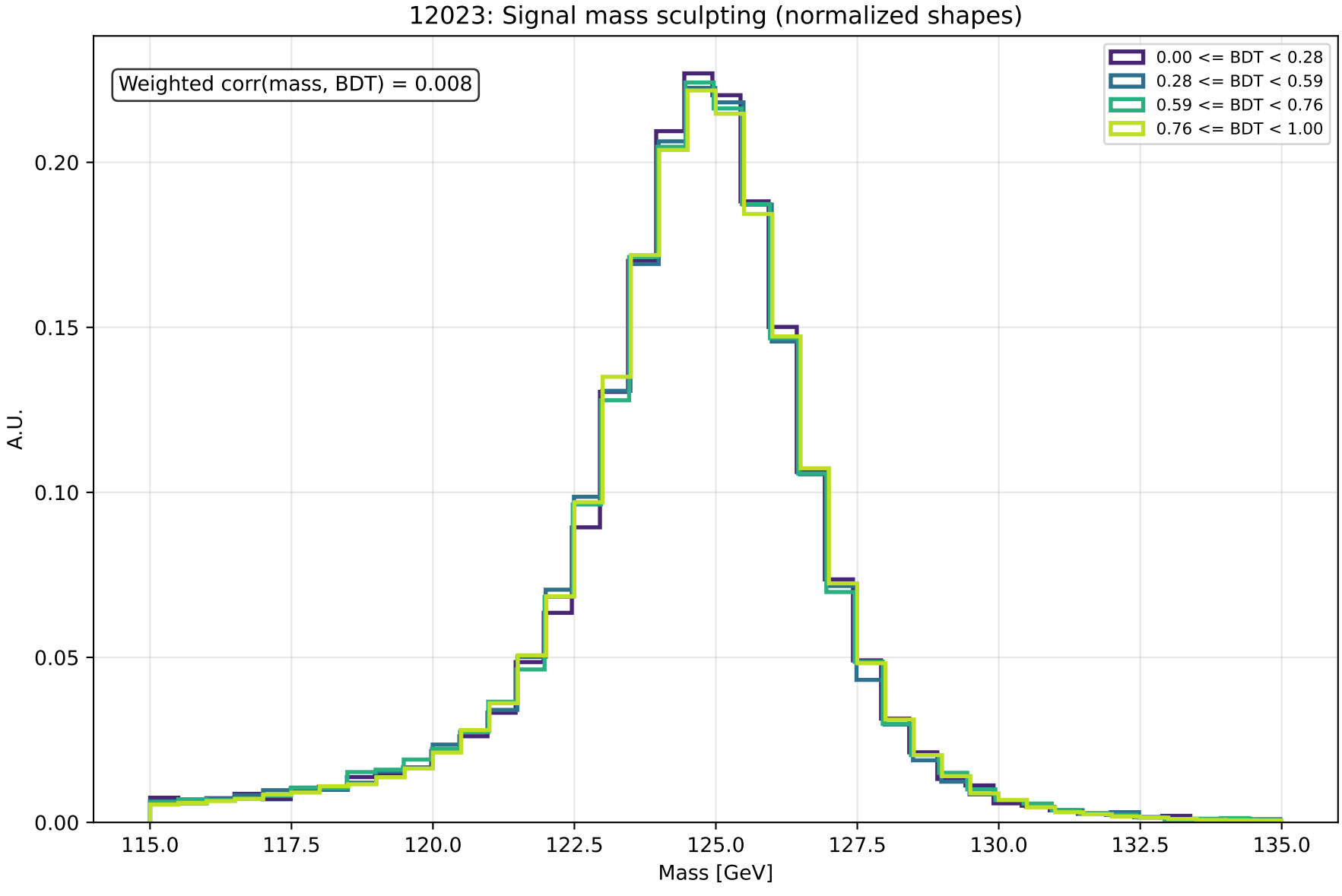
SumExp	LandaUxBernstein	0.0000	0.0000	-1.0000	FAIL
SumExp	BWZxBernstein	1.9150	2.0548	0.9150	FAIL
SumExp	BWZGamma	7.1283	1.0422	6.1283	FAIL
PowerLaw	BWZRedux	1.8967	1.9034	0.8967	FAIL
PowerLaw	SumExp	9.5417	2.9271	8.5417	FAIL
PowerLaw	PowerLaw	2.3617	2.4358	1.3617	FAIL
PowerLaw	FEWZxBernstein	1.6650	2.3371	0.6650	FAIL
PowerLaw	LandaUxBernstein	0.0450	0.1781	-0.9550	FAIL
PowerLaw	BWZxBernstein	2.2867	3.5150	1.2867	FAIL
PowerLaw	BWZGamma	6.9533	3.3299	5.9533	FAIL
FEWZxBernstein	BWZRedux	1.6667	2.0977	0.6667	FAIL
FEWZxBernstein	SumExp	10.2767	2.9023	9.2767	FAIL
FEWZxBernstein	PowerLaw	2.0917	2.5535	1.0917	FAIL
FEWZxBernstein	FEWZxBernstein	1.3450	1.9373	0.3450	FAIL
FEWZxBernstein	LandaUxBernstein	0.0450	0.2423	-0.9550	FAIL
FEWZxBernstein	BWZxBernstein	1.6100	2.2089	0.6100	FAIL
FEWZxBernstein	BWZGamma	6.7917	3.7269	5.7917	FAIL
LandaUxBernstein	BWZRedux	453.8433	3.8350	452.8433	FAIL
LandaUxBernstein	SumExp	894.6117	4.1153	893.6117	FAIL
LandaUxBernstein	PowerLaw	580.4767	20.0493	579.4767	FAIL
LandaUxBernstein	FEWZxBernstein	737.2417	13.0836	736.2417	FAIL
LandaUxBernstein	LandaUxBernstein	0.9367	0.8873	-0.0633	PASS
LandaUxBernstein	BWZxBernstein	809.3967	9.7109	808.3967	FAIL
LandaUxBernstein	BWZGamma	907.3083	8.5069	906.3083	FAIL
BWZxBernstein	BWZRedux	4.1350	3.1174	3.1350	FAIL
BWZxBernstein	SumExp	7.5650	4.1541	6.5650	FAIL

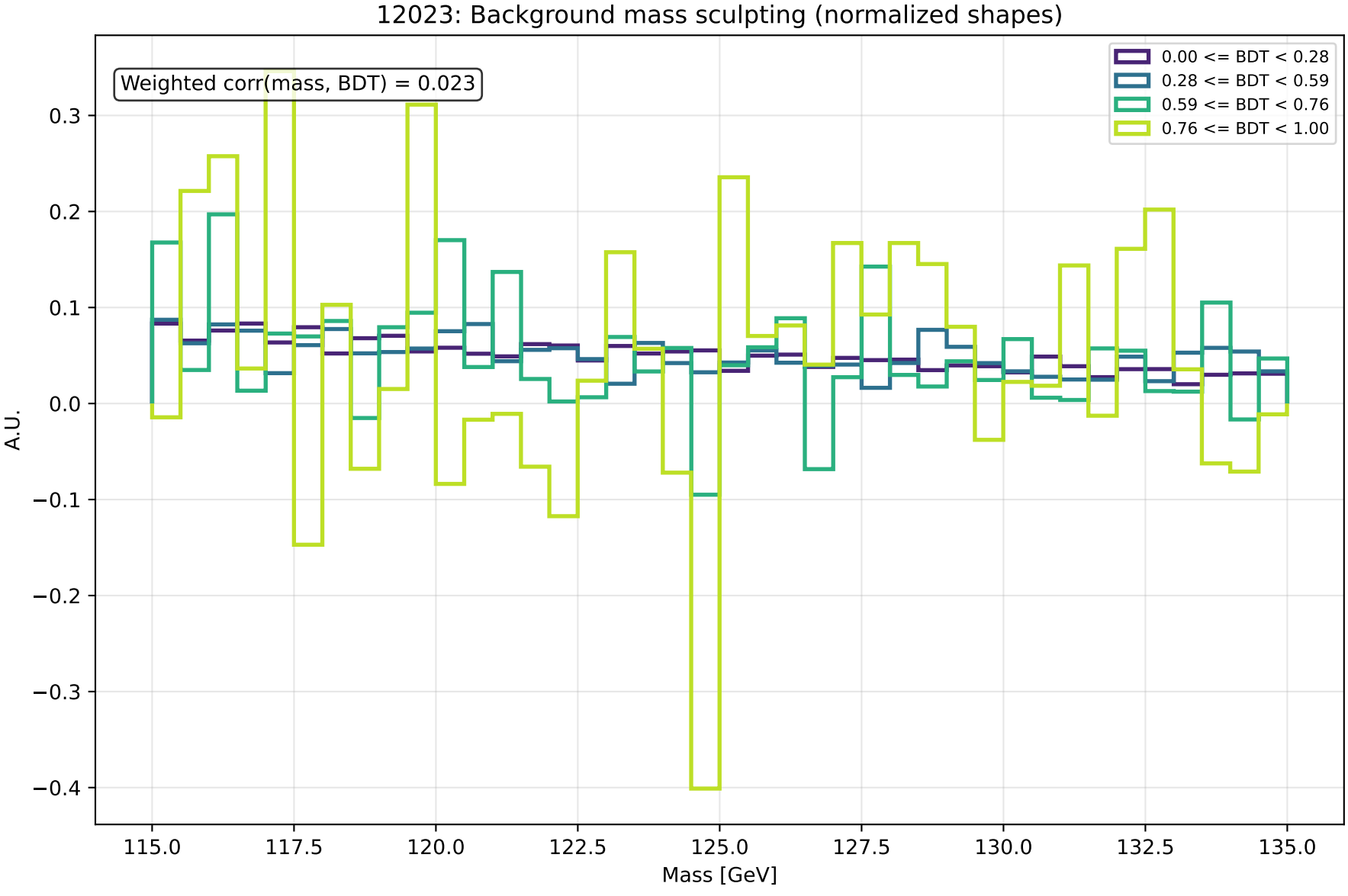
12023: Expected yield per optimized BDT bin (115-135 GeV)
S total=54.31, B total=81164.71

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.283	4.308	7.9%	100.0%	65419.940	0.000669
0.283-0.594	6.529	12.0%	92.1%	12651.131	0.00318
0.594-0.756	5.876	10.8%	80.0%	2300.436	0.0141
0.756-1.000	37.600	69.2%	69.2%	793.205	0.0474

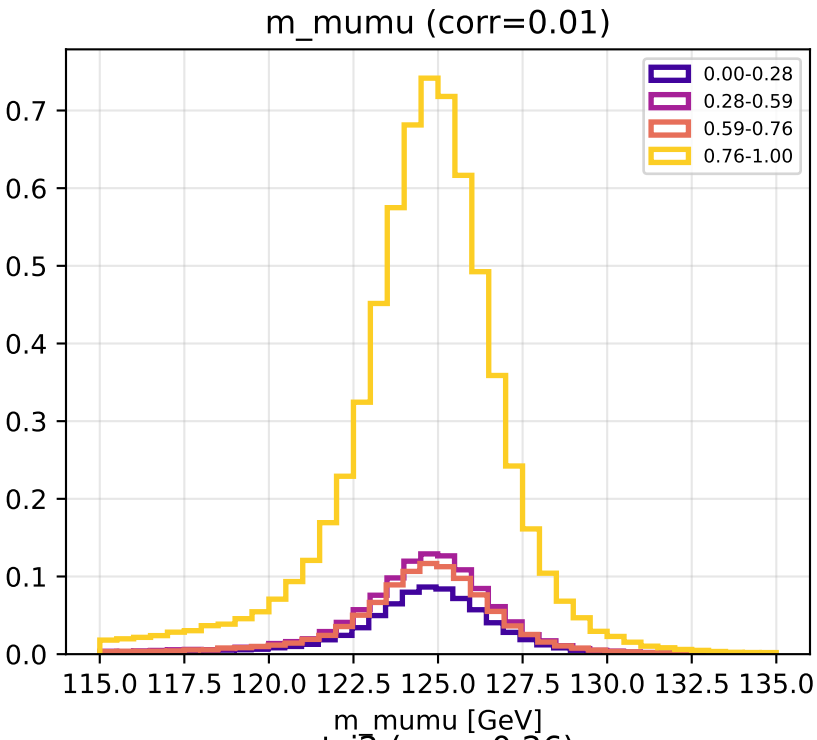

12023: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.28 4.31 124.48 2.45
0.28-0.59 6.53 124.47 2.45
0.59-0.76 5.88 124.47 2.46
0.76-1.00 37.60 124.52 2.41
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

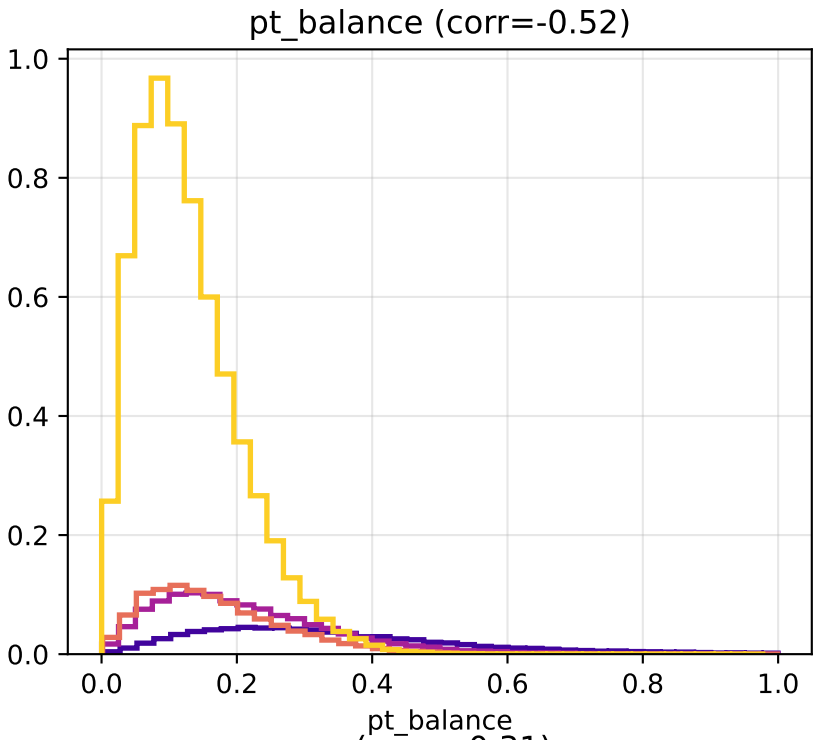

12023: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.28 65419.94 123.40 5.63
0.28-0.59 12651.13 123.88 5.91
0.59-0.76 2300.44 122.77 6.02
0.76-1.00 793.20 123.84 6.38
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
2500
2000
1500
1000
500
0
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

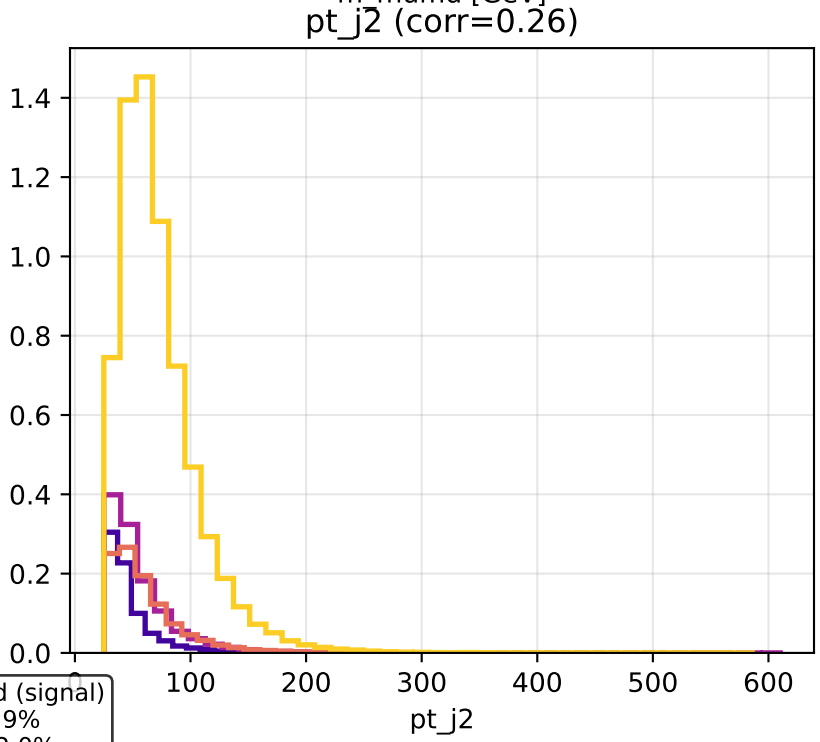

12023: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.008
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

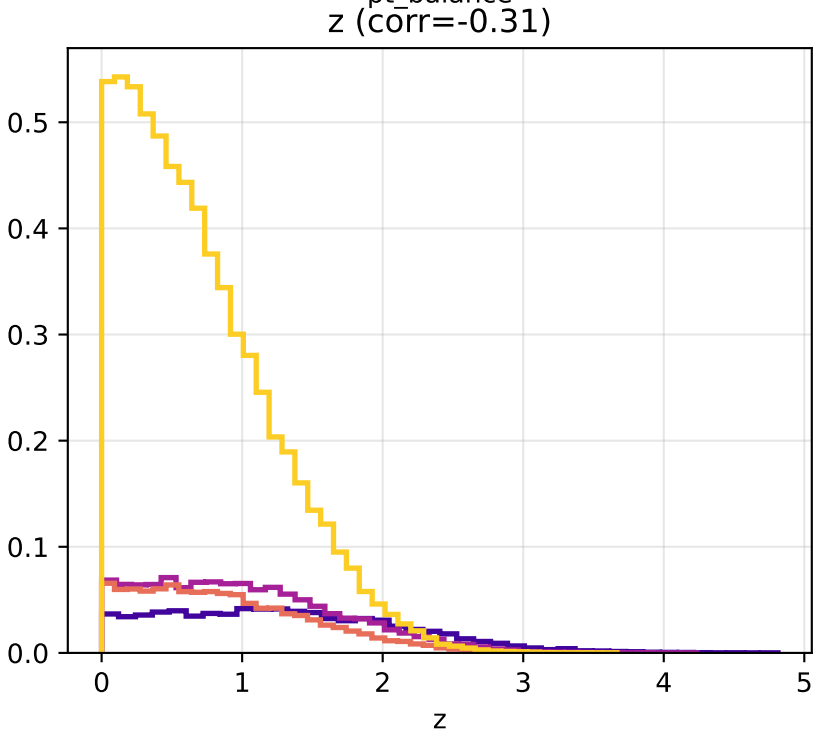

12023: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.023
0.00 <= BDT < 0.28
0.28 <= BDT < 0.59
0.59 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

12023: Signal distributions by BDT bin

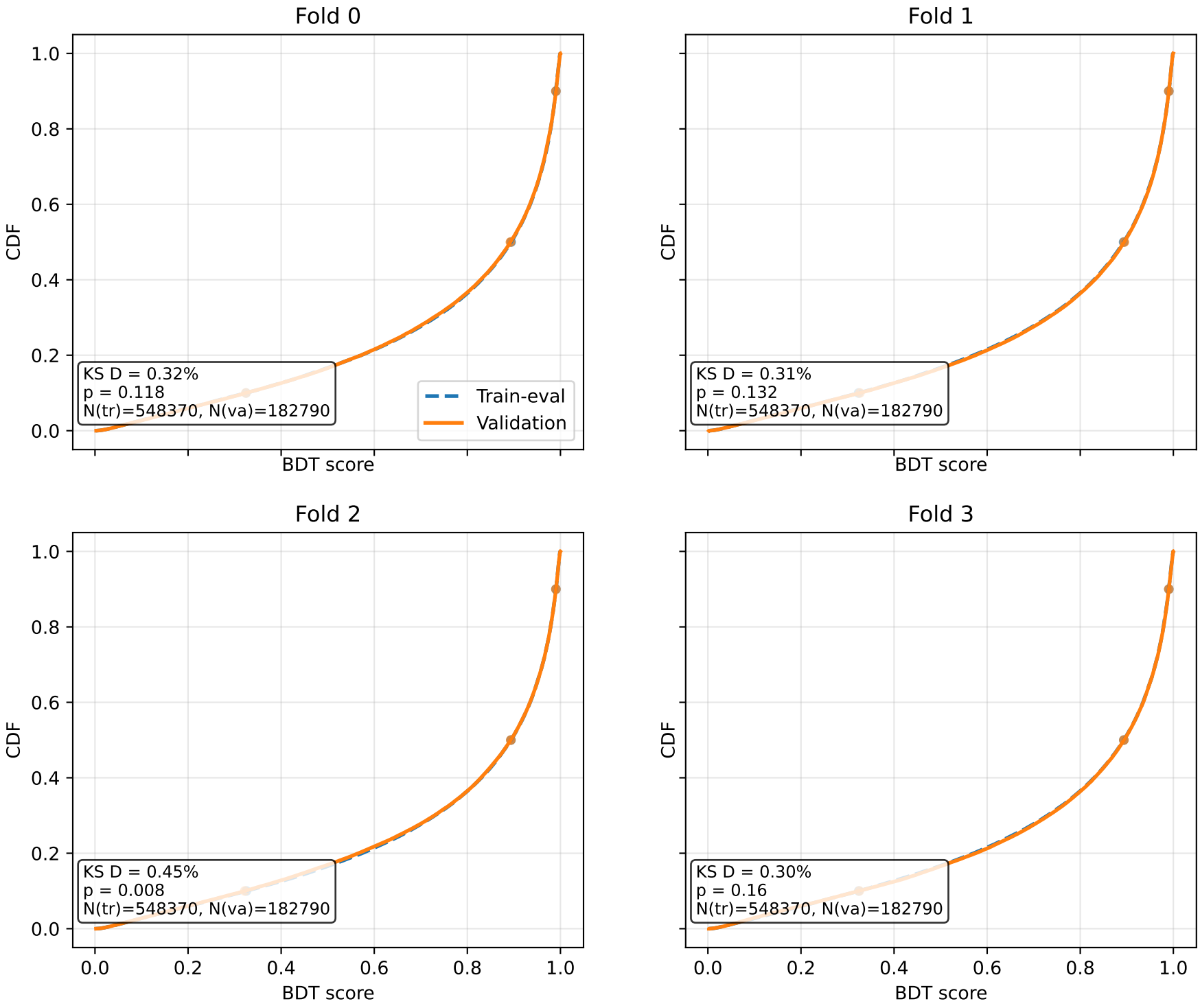








BDT bin yield (signal)
0.00-0.28: 7.9%
0.28-0.59: 12.0%
0.59-0.76: 10.8%
0.76-1.00: 69.2%


12023: CDF (train-eval vs validation)
Fold 0
KS D = 0.32%
p = 0.118
N(tr)=548370, N(va)=182790
Train-eval
Validation
Fold 1
KS D = 0.31%
p = 0.132
N(tr)=548370, N(va)=182790
Fold 2
KS D = 0.45%
p = 0.008
N(tr)=548370, N(va)=182790
Fold 3
KS D = 0.30%
p = 0.16
N(tr)=548370, N(va)=182790
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

12023: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.32%	182790	0.001	0.20%	FALSE
1	0.31%	182790	0.001	0.20%	FALSE
2	0.45%	182790	0.001	0.20%	FALSE
3	0.30%	182790	0.001	0.20%	FALSE

12023: Top-10 feature importance (gain)
12023_featureimportance_DY.txt

Feature	Gain %
pt_balance	15.97%
pt_j2	9.51%
z	8.51%
dphi_mumu_jj	8.17%
deta_jj	8.07%
pt_mumu	7.81%
pt_j1	7.06%
njets	6.03%
jet1_eta	4.01%
mu2_pt_over_m	2.94%